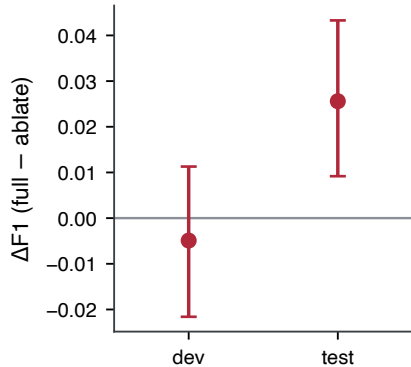
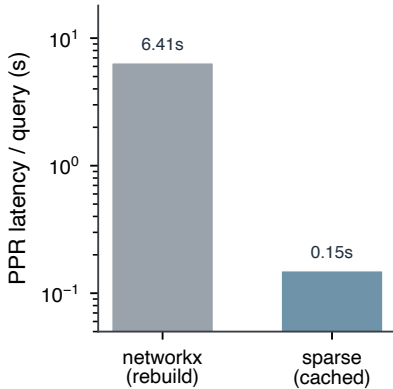


**(a) RAPTOR flips sign**



**(b) Naive vs. cached PPR**



**(c) Cross-doc: bug hid it**

